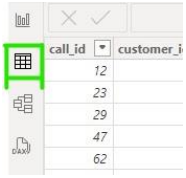
Lab 09 – Managing Relationships

In the following lab, you will open a Power BI file and assess the Relationships between the Tables. You will create some new Table Relationships and troubleshoot a Relationship issue.

Exercise 1 – Open a Power BI file and examine the Table relationships

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab09** folder. Double-click the **QAPBID Lab 9 - Start.pbix** file to open it.
3. In the left panel, select **Model** view. Review the Table Relationships.
4. The Model currently has:
    a) 8 tables - some of which are not related to any other tables.
    b) The three unrelated tables could be related to the other tables.
    c) Most of the existing relationships are 1:M.
    d) The relationship between Products and Product Costs is 1:1.
5. In the left panel, select **Table** view. Review the content of the Tables.



6. Using the **Fields** pane, select the **Category Managers** table. The Category column holds four values (Bikes, Clothing, Components and Accessories).
7. Select the **ProductCategory** table. The Product Category column holds five values (Bikes, Clothing, Components, Accessories and No category). This matches the Category Manager table values and it should be a 1:1 relationship. The No category value does not have a Manager assigned.
8. Select the **Region Managers** table. The Region column has seven country names.
9. Select the **Territories** table. The Country column has the same seven distinct values, but United States is repeated. This could be a 1:M relationship. The Territory column holds a list of territories within each country, although currently only the United States is split in this way.
10. Select the **Sales** table. The Territory column has the same content as the Territories table Territory column. This should be a 1:M relationship as the Sales table holds multiple sales for each territory.

Exercise 2 – Relate Category Managers to Categories

1. Select **Model** view.
2. Click and drag the **Category Managers** table to a position that is close to the **ProductCategory** table.
3. Click and drag the **Category** field from the **Category Managers** table onto the **Product Category** field in the **ProductCategory** table. When you release the mouse button, a relationship will be created. Power BI has determined the relationship is 1:1.
4. Position the **Territories** Table close to the **Sales** Table. Click and drag the **Territory** field from the **Territories** table onto the **Territory** field in the **Sales** table. Power BI has determined the relationship is 1:M.
5. Click and drag the **Region** field from the **Region Managers** table onto the **Country** field in the **Territories** table. Power BI has determined the relationship is 1:M.
6. **Double-click** on the relationship between **ProductCategory** and **Category Managers** to open the Edit relationship dialog box. The cardinality is set to One to One (1:1) and the cross-filter direction is set to both. Select **Cancel** to close the window.

Exercise 3 – Changing requirements

From: Sales Dept Manager
To: Power BI Developer
Subject: Staffing changes

Content: The United States region will now be managed by 3 Managers.

There is no territorial split at present, with all managers being able to sell into all territories.

1. The changes described above mean that the relationship between **Region Managers** and **Territories** should be a many-to-many (M:M) relationship. This is currently a 1:M relationship.

2. Using **Windows Explorer**, access the **C:\Coursefiles\QAPBIDESK\lab09** folder and rename **RegionManagers.txt** to **RegionManagersOLD.txt**.

3. In the same folder, rename **RegionManagers_NewStructure.txt** to **RegionManagers.txt**.

4. In **Power BI Desktop**, click the **Home** tab and click **Refresh.** Notice that the refresh fails as duplicate 'United States' values have been found on the 1 side of the 1:M relationship. The relationship would need to be changed to M:M before refreshing the data.

5. **Double-click** on the relationship between **Territories** and **Region Managers** to open the **Edit relationship** window. The **Cardinality** is set to Many-to-One (M:1). This should be changed to a **Many-to-Many** (M:M)

   relationship. Also change the **Cross filter direction** to **Both**. Select **OK** to save the changes.

6. **Refresh** the data again. As the relationship is correct, it works.

7. Click **File**, then **Save**.

8. Close Power BI Desktop.

Result: You assessed some Table Relationships, created some new Relationships, and resolved a Table Relationship issue.

Lab: 10 Customising your Data Model

In the following lab, you will look at how a Data Model can be customised to display your data in a more logical and presentable manner. You will display the data utilising the Table, Map and Matrix visuals.

Exercise 1 – Open a Power BI file and examine the Table relationships

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab10** folder. Double-click the
3. **QAPBID Lab 10 Start.pbix** file to open it.
4. Select **Model** view.
5. The Model currently has:
    a) 9 tables - one of which **Standard Cost (from Excel)** is not related to any other table.
    b) All existing relationships are 1:M.
    c) The **Standard Cost** table has been used to merge the **Standard Cost** values into the **Product** table and can be hidden.
6. Select **Data** view and review the Table data.

Exercise 2 – Sort the Months by Month number

1. Select **Report** view.
2. On the **Fields** pane, expand the **Order Date** table. Click and drag the **MonthTitle** field onto the report canvas. As the **MonthTitle** field has a **Text** data type, the default visual is a **Table** and the months are listed in alphabetic order - by default.
3. Select **Data** view.
4. From the **Fields** pane, expand **Order Date** and select the **MonthTitle** column. With the **MonthTitle** column **(Column Tools)** highlighted, on the ribbon, click the **Sort by column dropdown** and select **CalendarMonth**. The **MonthTitle** is now sorted by the **CalendarMonth** number rather than alphabetically.
5. Select **Report** view. The visual now displays the **MonthTitle** in the correct order.

Exercise 3 – Formatting the column values

1. Ensure that the **MonthTitle** table visual is selected. From the **Fields** pane, expand the **Reseller Sales** table. Select the **OrderQuantity** and **SalesAmount** fields to add them to the Visual.

2. Select **Data** view. **(Can also be done when selecting column in Report View)**

3. Expand the **Reseller Sales** table and select the **SalesAmount** column.

4. On the **Ribbon**, change the **Format** properties from **General** to **Currency** and **$** to

5. **£ English (United Kingdom).** Confirm that the **Comma** symbol is selected and change **Auto** to **2**. Scroll down to review the changes to the sales values.

6. Select **Report** view.

7. The visual will now show the formatted numbers.

Exercise 4 – Using the 'Data Category' option

1. Select **Data** view.

2. From the **Fields** pane, expand the **Geography** table. Select the **City** column.

3. On the **Ribbon**, change the **Data Category** value to **City**. Notice that the property change has added an **icon** to the **City field** in the **Fields** pane (small globe symbol).

4. Select the **RegionName** column.

5. Change its **Data Category** to **Country** (NOT County).

6. Select **Report** view.

7. From the **Fields** pane, expand the **Geography** table. Click and drag the **RegionName** field onto the report canvas. Each country with data now has a bubble added. You may need to resize the map to display every country.

8. From the **Reseller Sales** table, click and drag the **OrderQuantity** field onto the Map visual. The bubbles change size relative to the quantity of Orders.

Exercise 5 – Add Continents

1. Select **Data** view.
2. Verify that the **RegionName** column in the **Geography** table is still selected.
3. Select **Data groups,** then **New data groups** to add a group based on the column.
4. In the **Groups** window, change the **Name** to **Continents**.
5. From **Ungrouped values**, select **Canada** - hold down the **Ctrl key** – and select **United States.** Click the **Group** button. On the right, change the name of the new group to **North America (Double Click).**
6. Next, select **France**, **Germany,** and **United Kingdom** together. Click **Group**. Change the name of this group to **Europe**.
7. Select **Australia.** Click **Group**. Change the group name to **Oceania**.
8. Click **OK**. The **Continents** data group column is added to the **Geography** table.

Exercise 6 – Create a hierarchy for Geography

1. Select **Model** view.
2. From the **Fields** pane**, right-click** the **Continent** field in the **Geography** table, select **Create hierarchy**. The new hierarchy appears close to the bottom of the **Properties** pane (on the left).
3. In the **'Select a column to add level'** dropdown, select the **RegionName** field to add it to the hierarchy list.
4. Repeat this for the **TerritoryName**, **StateProvinceName**, and **City** fields.
5. If the hierarchy order is correct, click '**Apply Level Changes'**.
6. At the top of the **Properties** pane, change the **Name** of the hierarchy to **World**.
7. Select **Report** view.
8. Add a **Matrix** visual to the report page.
9. From the **Geography** table, click and drag the **World** hierarchy onto the **Matrix** visual.
10. From **Reseller Sales** table, click and drag the **OrderQuantity** field onto the **Matrix** visual.
11. Click the **Format visual** icon. Under **Row headers**, change the **+/- icons** to **On**. Expand its settings and increase the **Size** to **10px**.

    Above the visual, select the **Expand all down one level in the hierarchy** icon.(Forked arrow) The continents and countries now display. **Keep selecting the same icon**. The table expands to display data at the territory, state province and city levels.

    Use the **Drill Up** arrow to rise up through the levels.

Exercise 7 – Hiding content

1. Within the **Fields** pane, select the **Reseller Sales** table. Some of the columns shown would not normally be used by the report designer. **Right-click** on the each of the fields starting with **SK.** Select **Hide**.

2. The **Standard Cost (from Excel)** table is not required to be shown to the report user, so this should be hidden. **Right-click** the **Standard Cost** table, select **Hide**.

3. Click **File,** then **Save**.

4. Close Power BI Desktop.

MonthTitle	OrderQuantity	SalesAmount
January	1,927	£434,297.74
February	3,939	£1,304,942.08
March	2,583	£899,786.35
April	2,725	£567,946.39
May	5,814	£1,625,101.67
June	3,837	£1,193,981.20
July	2,107	£483,203.57
August	4,011	£1,338,045.41
September	1,839	£655,520.77
October	846	£349,574.54
November	1,242	£311,739.93
December	4,189	£1,743,502.37
Total	**35,059**	**£10,907,642.02**

Continent	OrderQuantity
Europe	**35,059**
France	**14,348**
France	**14,348**
Charente-Maritime	**1,523**
Saint Ouen	1,523
Essonne	**381**
Morangis	372
Verrieres Le Buisson	9
Garonne (Haute)	**578**
Colomiers	578
Hauts de Seine	**2,369**
Colombes	1,317
Courbevoie	766
Paris La Defense	53
Sèvres	147
Suresnes	86
Loiret	**3,020**
Orleans	3,020
Nord	**158**
Croix	147
Total	214,516

Result: You customised a Data Model to display the data in a more presentable format. You presented the data utilising the Table, Map and Matrix visuals.

Lab 12: Tabular Visuals

In the following lab, you will configure some report settings and create some tabular style report visuals.

Exercise 1 – Wallpaper & page setup

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab12** folder. Double-click the
3. **QAPBID Lab 12 Start.pbix** file to open it.
4. Select **Report** view.
5. **Click the background** of the **report canvas**. In the **Visualizations** panel, click the **Format** icon. Expand **Page information** and change the page name to **Regional Sales.**
6. Expand **Wallpaper**, under **Image**, browse to and select the wallpaper image

   ...\lab12\qa wallpaper.png. Set **Image Fit** to **Fill**.

Exercise 2 – Sales by Quarter (Table Visual)

7. Select the **Build** icon. Click the **Table** Visual.
8. From the **Fields** pane, drag the following fields onto the table:
   - **Order Date** \ Quarter
   - **Web Sales** \ Web Sales Amount
   - **Web Sales** \ Web Quantity
   - **Web Sales** \ Web Sales %Change YonY
9. **Resize the table** to show all quarters. Position the table near the top left of the page.
10. Ensure that the **table is selected** and click the **Format** icon.
11. Under **Cell elements**, set the **Series** column to **Web Quantity.** Set **Data bars** to **On**. Click the ***fx* button.** In the dialog window, you can change colour settings and minimum/maximum limits. To keep with the colour theme, set the **Positive** colour to a **mid blue**. Click **OK**.
12. Under **Cell elements**, change the **Series** column to **Web Sales Amount.** Set **Background color** to **On**. Click the ***fx* button.**

    In the **Background color** dialog window, select the following options:

    Format style = Rules

    Based on field = **Web Sales** \ Web Sales %Change YonY

    Rule 1: If value >= to -1 and is < -0.2 then (pale red). Click + **New Rule Rule 2:** If value >= to -0.2 and is < 0.2 then (pale yellow). Click + **New Rule**
    Rule 3: If value >= to 0.2 and is < 100 then (pale green).

    Set all of the above rules to **Number** values. Click **OK**.

The background colours will be displayed in the **Web Sales Amount** column using the **Web Sales %Change YonY** column as the reference.

Exercise 3 – Sales by Geography and Category (Matrix Visual)

1. Add a **Matrix** visual to the page positioned to the right of the existing table. Resize the Matrix to use all remaining space on the page (without covering the logo).

2. From the **Fields** pane, drag the following fields to their respective **buckets**:

   Geography \ World hierarchy to **Rows** box **Product** \ CategoryName to **Columns** box
   Web Sales \ Web Quantity to **Values** box

3. Ensure that the Matrix is selected (and displaying data at the 'top level'). Click the Format icon.

4. Under **Row headers**, set the **+/- icons** to **On**. In the **Matrix** each Country now displays a **+** symbol to allow easier expansion of the hierarchy. This will change to a **–** symbol to collapse the expanded Country. To make the icons easier to click, change the icon **Size** to **10**.

5. **Save** the Power BI Desktop file.

6. Close Power BI Desktop.

Quarter	Web Sales Amount	Web Quantity	Web Sales %Change YonY
Q1 2016	£914,243.38	281	
Q2 2016	£1,538,486.71	479	
Q3 2016	£1,943,518.75	607	
Q4 2016	£1,991,419.40	622	
Q1 2017	£1,679,427.93	725	83.70 %
Q2 2017	£1,114,786.12	625	-27.54 %
Q3 2017	£1,530,018.83	859	-21.28 %
Q4 2017	£1,541,061.97	892	-22.61 %
Q1 2018	£2,225,246.09	5155	32.50 %
Q2 2018	£3,244,665.26	12146	191.06 %
Q3 2018	£4,560,193.00	14639	198.05 %
Q4 2018	£4,955,585.18	15255	221.57 %
Q1 2019	£5,439,678.21	19673	144.45 %
Total	**£32,678,330.83**	**71958**	**19.97 %**

Continent	Accessories	Bikes	Clothing	Total
⊟ **Europe**	**12921**	**5567**	**3031**	**21519**
⊟ **France**	**4020**	**1638**	**925**	**6583**
⊞ France	4020	1638	925	**6583**
⊟ **Germany**	**3996**	**1824**	**913**	**6733**
⊞ Germany	3996	1824	913	**6733**
⊞ **United Kingdom**	**4905**	**2105**	**1193**	**8203**
⊟ **North America**	**22337**	**6648**	**5779**	**34764**
⊞ **Canada**	**6511**	**1054**	**1640**	**9205**
⊟ **United States**	**15826**	**5594**	**4139**	**25559**
⊞ Central	21	2	5	**28**
⊞ Northeast	27	3	5	**35**
Total	**43706**	**17177**	**11075**	**71958**

Result: You customised some report settings, including a background image, and created some Tabular style report visuals.

Lab: 13 Chart Visuals

In the following lab, you will configure a range of different charts which highlight the versatility of the available visuals. You will also fine-tune and customise some of their settings.

Exercise 1 – Stacked Column Chart

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab13** folder. Double-click the
3. **QAPBID Lab 13 Start.pbix** file to open it.
4. Select **Report** view.
5. At the bottom of the Report, **click the + symbol** to add a new report page.
6. **Double-click** the new **Page tab** and rename it **Web Sales Charts.**
7. Select a **Stacked Column Chart** from the visuals. **Resize the chart** to approximately a quarter of the page. **Position it in the top left quadrant**.
8. From the **Fields** pane, drag the following fields to the specified field bucket:
   - **Product** \ SubcategoryName to **X-axis**
   - **Web Sales** \ Web Quantity to **Y-axis**
   - **Product** \ CategoryName to **Legend**
9. Ensure that the chart is selected. Click the **Format** icon, under **Columns**, select various **shades of Blue** for each **category**.

Exercise 2 – Line and Clustered Column Chart

1. Click the page background to deselect the Column chart.
2. Select a **Line and Clustered Column Chart** from the Visuals. **Resize the chart** and place it in the **top right quadrant**.
3. From the **Fields** pane, drag the following **fields** into their respective buckets on the **Visualizations** pane:
   - **Order Date** \ Month to **X-axis**
   - **Web Sales** \ Web Sales Amount to **Column y-axis**
   - **Web Sales** \ Web Margin % to **Line y-axis**
4. Select the **Order Date** table. Drag the **CalendarYear** field to the **Filters** pane (which is to the left of the **Visualizations** pane). Add it to the 'Filters on this page' bucket. Change the Filter type to **Basic filtering**. Select only **2017** and **2018**. All charts will now only display values for these selected years.
5. Ensure that the **Line and Clustered Column Chart** is selected. Select the **Format** icon. Under **Columns**, select a **Blue** colour as the **Default**.
6. Set the **Secondary y-axis** to **On.** Set **Minimum** to **0** and **Maximum** to **2**.

Exercise 3 – Waterfall Chart

1. Click the page background to deselect the Column and Line chart.
2. Select a **Waterfall chart** from the visuals. Resize the chart and place it in the **bottom left quadrant** of the page.
3. From the **Fields** pane, drag the following **fields** into their respective buckets on the **Visualizations** pane: • **Order Date** \ Month to **Category**
   - **Web Sales** \ Web Sales Change MonM to **Y-axis**

Exercise 4 – Pie Chart

1. Click the page background to deselect the Waterfall chart.
2. Select a **Pie chart** from the visuals. Place the chart in the **bottom right quadrant** of the page.
3. From the **Fields** pane, drag the following **fields** into their respective buckets on the **Visualizations** pane:
   - **Product** \ CategoryName to **Legend**
   - **Web Sales** \ Web Quantity to **Values**
4. Ensure that the chart is selected. Select the **Format** icon, under **Slices**, change the category colours to various shades of **Blue**.
5. Under **Detail labels**, change the **Label contents** to **All detail labels**.

Exercise 4 – Interactivity

1. To test the interactivity of the charts, click on the **Clothing** slice of the **Pie chart**. The selection of the pie slice will either highlight or filter the other charts, dependent upon the chart type.
2. Explore chart filtering by clicking each chart in turn.
3. **Save** the Power BI Desktop file.
4. Close Power BI Desktop.

Result: You created a range of chart visuals and customised some of their relevant settings.

Lab: 14 Map Visuals

In the following lab, you will configure some Map charts which allow you to visually display Sales data relative to city locations.

Exercise 1 – Shape Map Chart

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab14** folder. Double-click the **QAPBID Lab 14 Start.pbix** file to open it.
3. Select **Report** view.
4. At the bottom of the Report, **click the + symbol** to add a new report page.
5. **Double-click** the new **Page tab** and rename it **UK Sales.**
6. From the **Fields** pane, expand the **Geography** table. Click and drag the **RegionName** field to the **Filters** pane bucket (Filters on this page). Under **Basic filtering**, select the **United Kingdom**.
7. Click and drag the **StateProvinceName** field onto the **Report canvas**.
   Resize the Map chart until it takes up the **left-hand third of the page**.
8. With the **Map visual** selected, change the visual to a **Shape Map**. By default, US States are displayed for the Shape map. Click the **Format** icon, expand **Map settings**. In the **Map type** dropdown, choose **Custom map** (at the very bottom). In **Add a map type**, browse to the **...\lab14 folder** and select the **uk-counties.json** file. Click **Open**. The map will change and display UK Counties.
9. On the **Visualizations pane**, click the **Build visual** icon (to the left of the **Format** icon). From the **Fields** pane, expand the **Web Sales** table. Click and drag the **Web Quantity** field into the **Color saturation** bucket. The shapes will change colour to show the relative sales quantities.
10. To highlight the counties with both sales and no sales using your own colour styling, move the **Web Quantity** field from the **Color saturation** bucket to the **Tooltips** bucket just below.
11. Select the **Format** icon. Under **Fill colors**, click the ***fx* button**. In the **Field dropdown** (What field should we base this on?), choose **Web Sales / Web Quantity.** Choose your own **Minimum** and **Maximum** colours.
12. To display the counties with no sales, under **Blank areas**, change the **Color** to **white.** Under **Border**, set the **Color** to **black**.

Exercise 2 – Map Chart

1. Deselect the Shape Map.
2. From the Fields pane in the **Geography** table, drag the **City** field onto the report page to the right of the Shape map. **Resize the Map visual** to fill the page area. The cities in the Geography table will be displayed on the map.
3. From the **Web Sales** table, drag the **Web Quantity** field to the **Bubble size** bucket in the **Visualizations pane**. The bubble sizes have changed to show the relative Sales quantities.
4. From the **Product** table, drag the **CategoryName** field to the **Legend** bucket. The bubbles are displayed as Pie charts.
5. Click the **Format** icon, expand **Map settings** and change the **Style** to **Grayscale**.

Exercise 3 – Interactivity

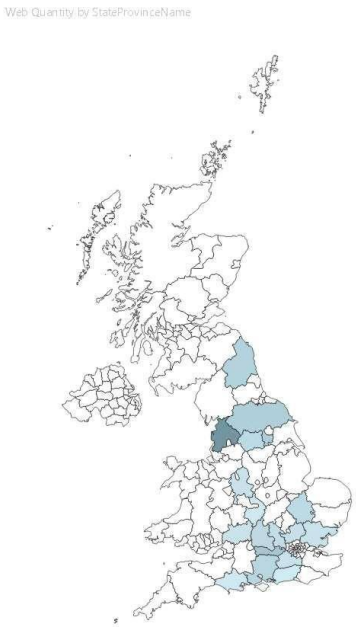
1. To test the interactivity, select a **County** in the **Shape map.** This will zoom into the cities within the County on the right.
2. Explore chart filtering by clicking each chart in turn.

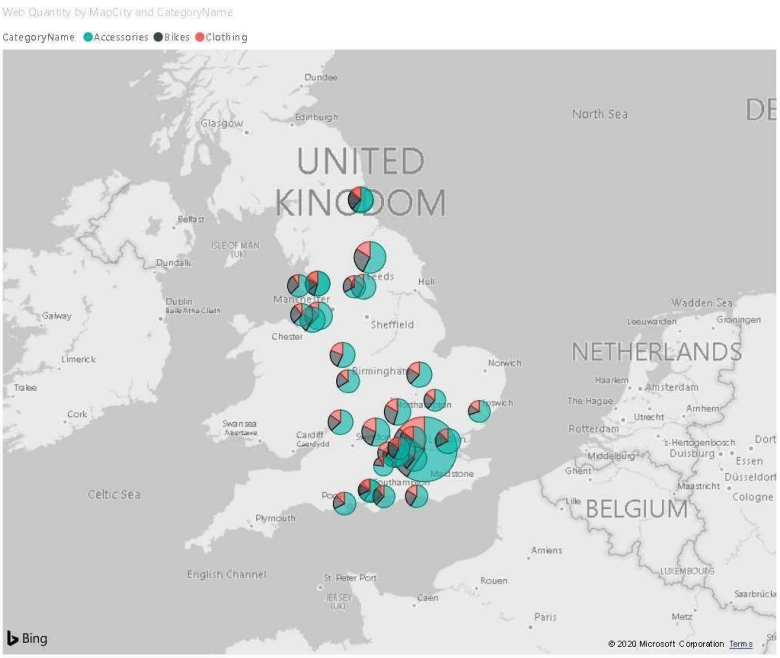
Additional Tip: If you zoom out, you will notice that some similar city names have been unintentionally highlighted in the USA; **Cambridge** and **Peterborough**. To remove them from the selection, in the **Geography** table, deselect the **City** field, and instead drag the **MapCity** field to the **location** bucket instead.

With the **MapCity** field selected, on the **Ribbon**, change the **Data category** to **Place**.

According to Microsoft, Bing can display location data more accurately using the 'place' data category when a location address exists in a single field. E.G., 1 Microsoft Way, Redmond Washington 98052 (which is how it appears in the **MapCity** column).

3. **Save** the Power BI Desktop file.
4. Close Power BI Desktop.





Result: You have created some Map charts which can visually provide an overview of Sales data by city location.

Lab: 15 Adding Slicers

In the following lab, you will add some Slicers which allow you to focus in on 'slices of data'. You will also examine how interactive updates between the Visuals can be restricted and controlled.

Exercise 1 – Add a Region Slicer

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab15** folder. Double-click the
3. **QAPBID Lab 15 Start.pbix** file to open it.
4. Select **Report** view.
5. From the **Visualizations** pane, select the **Slicer** visual.
6. From the **Fields** pane, expand the **Geography** table. Drag the **RegionName** field onto the **Slicer**.
7. **Resize the Slicer** so that the country names are clearly visible, but with no additional wasted space at the bottom. Position the Slicer in the '**Reseller Search Page'** space at the top. The **default view** for a text Slicer is a **List** and **single selection** only.
8. Select **Australia** within the **Slicer.** Notice that the other visuals on the page change appropriately.
9. To select multiple countries, hold down the **CTRL key** and select **Canada** as well.
10. Ensure the Slicer is selected. Click the **Format** icon on the **Visualizations** pane.
11. Expand **Slicer settings** and then **Selection**. Set **Show "Select all" option** to **On** and **Multi-select with CTRL** to **Off**. The **Slicer** visual now shows **small ticks** against Australia and Canada, plus the option to 'Select all'.
12. Choose **Germany** within the **Slicer** to add that to the selection. Notice that there is no need to hold the CTRL key down.
13. Change the selection on the **Slicer** to **France** only.
14. Hover over the **Slicer** and click the **small down arrow** (in the top right). Choose **Dropdown**. While the dropdown requires less screen space, the options are initially hidden.
15. Change the option back to **List**.
16. Change the selection to **Select All**.

Exercise 2 – Add a First Order Slicer

1. Make sure that no Visuals are selected. **Click** the **Slicer icon** to create another Slicer.
2. From the **Reseller** table, drag the **First Order** field onto the Slicer. If necessary, **resize** the **Slicer** width so that it matches the frame width.
3. Set the **First Order** range between **01/01/2018** and **09/04/2018**.
4. Hover over the **Slicer.** Click the **small down arrow** in the top right. Try the **Before** and **After** options. Take note of the changes to the data in the other visuals.
5. Change the setting to **Relative Date**. Set the **filter** to **Last 5 years**. Note the list of reseller orders changes to display only those in the last 5 years.
6. Set the **filter** to **Last 10 years**.
7. **Resize** the **Slicer** so that the date details are clearly visible, but with no additional wasted space at the bottom.

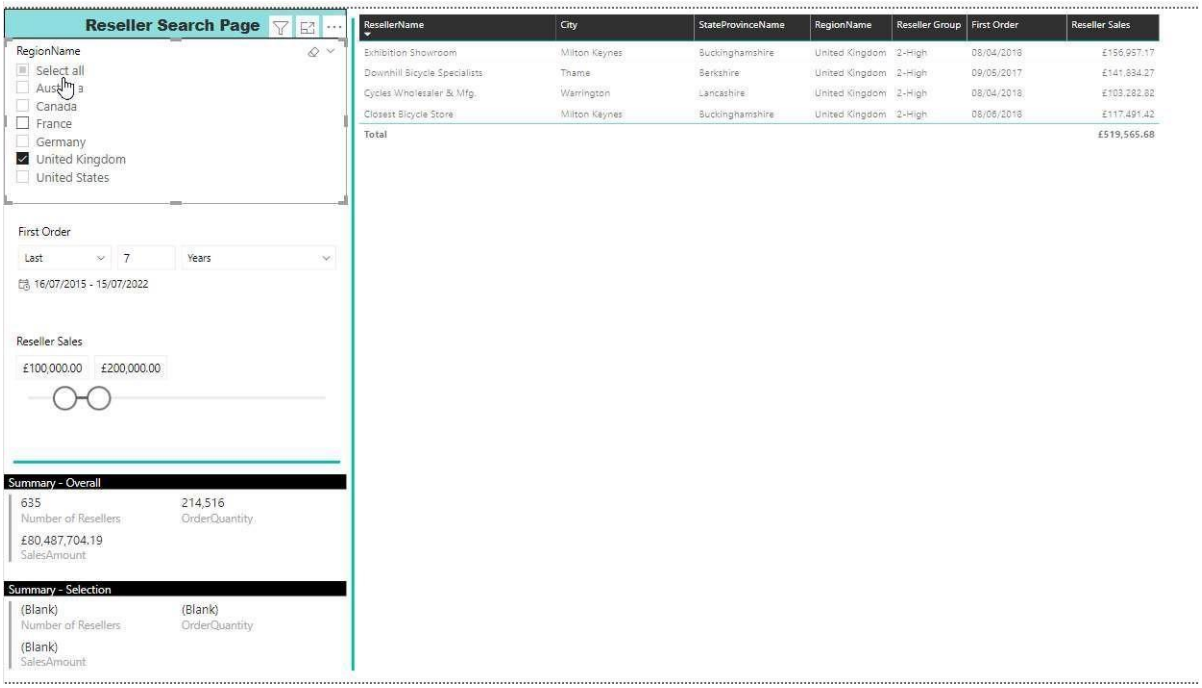
Exercise 3 – Add a Reseller Sales Slicer

1. Make sure that no Visuals are selected. **Click** the **Slicer icon** to create another Slicer.
2. From the **Reseller** table, drag the **Reseller Sales** field onto the Slicer. If necessary, **resize** the **Slicer** width so that it matches the frame width. As **Reseller Sales** is a column calculation, it can be used as a Slicer, whereas a calculated measure cannot.
3. Ensure that the **Reseller Sales field** is selected. From the **Column tools** ribbon, change the **Format** of the field to **Currency**, **£United Kingdom** and **decimal place setting to 2**.
4. Using the **Slicer,** set the **Reseller Sales** range between **£10,000** and **£15,000**.
5. Hover over the **Slicer.** Click the **small down arrow** in the top right. Notice the list of possible settings.
6. Use the **Slicer** to set the **Reseller Sales range** between **£0** and **£1,000,000**.
7. Resize the Slicer size to reduce wasted space.

Exercise 4 – Restrict filtering of the Summary – Overall Multi-row Card visual (using Edit Interactions)

There is a **Summary - Overall** Multi-row card visual (bottom left corner). We can stop this visual from filtering (updating) by turning its interactivity off if we choose.

1. To begin, select the RegionName slicer (at the top) and deselect all countries.
2. Click the **Format** tab on the **Ribbon**. Click **Edit Interactions** (this will toggle the feature on). New icons are displayed for each of the visuals. The icons represent **Filter** and **None**.
   - Ensure that the **RegionName slicer** is still selected, then click the **None** icon ('No entry sign') to the right of the **Summary – Overall** Multi-row Card.
   - Select the **First Order** slicer. Select the **None** icon on the **Summary – Overall** Multi-row card.
   - Select the **Reseller Sales** slicer. Select the **None** icon on the **Summary – Overall** Multi-row card.
   - Select the **Table** visual (to the right). Select the **None** icon on the **Summary - Overall** Multi-row card.
3. On the **Format** menu, click the **Edit Interactions** button to toggle it off.
4. Set the Slicer values as below:
   - **RegionName** = United Kingdom
   - **First Order** = Last 7 calendar years
   - **Reseller Sales** = £100,000 to £200,000
5. Notice how the **Summary – Overall** visual **does not interactively update**, yet the other visuals do. This is because its interactivity was set to **None** when the other visuals are updated.
6. **Save** the Power BI Desktop file.
7. Close Power BI Desktop.



Result: You have created some Slicers to help select and highlight the data you need to view. You also edited the visuals to restrict updates to the **Summary - Overall** visual.

Lab: 17 Adding Buttons and Bookmarks

In the following lab, you will create some buttons, that when clicked, will display different charts based on the defined bookmarks.

Exercise 1 – Create the Buttons

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab17** folder. Double-click the
3. **QAPBID Lab 17 Start.pbix** file to open it.
4. Select **Report** view.
5. Click the **Insert** menu tab, click **Buttons**, select **Blank**. Click and drag the new button to a position above the column chart, just to the right of the vertical line.
6. **Format** the button as below: **Button | Style | State** = Default

   Button | Style | Text = On **| Text** = Show Chart **| Font Color** = Black **Button | Style | Fill** = On **| Color** = Light Grey

   Button | Style | State = On hover

   Button | Style | Fill = On **| Color** = Mid Grey

7. Check your settings – when you hover over the button, its colour should change.
8. On the **Insert** Ribbon, click **Buttons**, select **Blank**. Move this new button just to the right of the **Show Chart** button.
9. **Format** the new button as below - and then test your settings as above:

   Button | Style | State = Default

   Button | Style | Text = On **| Text** = Show Matrix **| Font Color** = Black **Button | Style | Fill** = On **| Color** = Light Grey

   Button | Style | State = On hover

   Button | Style | Fill = On **| Color** = Mid Grey

Exercise 2 Create the Bookmarks

A bookmark represents *the objects that you want to display* when you click a specific button.

1. On the **View** menu tab, click the **Bookmarks** and **Selection** ribbon buttons to display the panes.

2. First, we'll display the objects that we want to see when we click the 'Show Chart' button.

   On the **Selection** pane, **click** the **eye symbol** to **HIDE** the **Show Chart** button and the **Matrix** chart. They should now be hidden on the canvas.

   Verify that only the **Show Matrix** button and **Column chart** are displayed.

3. On the **Bookmarks** pane, click **Add**.

4. Double-click **Bookmark 1** and rename it **Show Chart**.

5. **Click** the **ellipses (…)** on the right of the bookmark entry and **untick Data**. This will ensure that any filters selected will be kept when using the bookmark.

6. Next, the objects that we want displayed when we click the 'Show Matrix' button.

   On the **Selection** pane, **click** the **eye symbol** to redisplay the **Show Chart** button and **Matrix** chart. And then **HIDE** the **Show Matrix** button and **Column** chart.

7. On the **Bookmarks** pane, click **Add**.

8. Double-click **Bookmark 2** and rename it **Show Matrix**.

9. **Click** the **ellipses (…)** on the right and **untick Data**.

Exercise 3 – Link the Buttons to the Bookmarks

1. Select the **Show Chart** button on the canvas.

2. On the **Format button** pane, set **Action** to **On**. Set **Type** to **Bookmark.** And **Bookmark** to **Show Chart**.

3. Click the Canvas background to deselect the button.

4. On the **Selection** pane, **click** the **eye symbol** to display the **Show Matrix** button and **select** it**.**

5. On the **Format button** pane, set **Action** to **On**. Set **Type** to **Bookmark.** And **Bookmark** to **Show Matrix**.

6. Click the canvas background to deselect the button.

–

Exercise 4 Test the Buttons

1. Test the **Show Chart** button by holding down the **CTRL** key and **clicking the button**. The Column chart should display, along with the Show Matrix button.

2. Ctrl-click the **Show Matrix** button. The Matrix chart should display, along with the Show Chart button.

Exercise 5 – Positioning the Buttons and Charts Buttons

1. **Select** the **Show Chart** and **Show Matrix buttons individually**. Unhide them if necessary. Apply the settings below to both of them.

2. On the **Format button** pane, **General** tab:

   - **Properties | Size** ○

     Height = 40 ○ **Width**

     = 100

   - **Properties | Position**

     ○ **Horizontal** = 240 ○

     Vertical = 10 **Charts**

1. **Select** the **Matrix** and **Column charts individually**. Unhide them if necessary. Apply the settings below to both of them.

   On the **Format Visual** icon, **General** tab:

   - **Properties | Size** ○ **Height**

     = 600 ○ **Width =** 1000

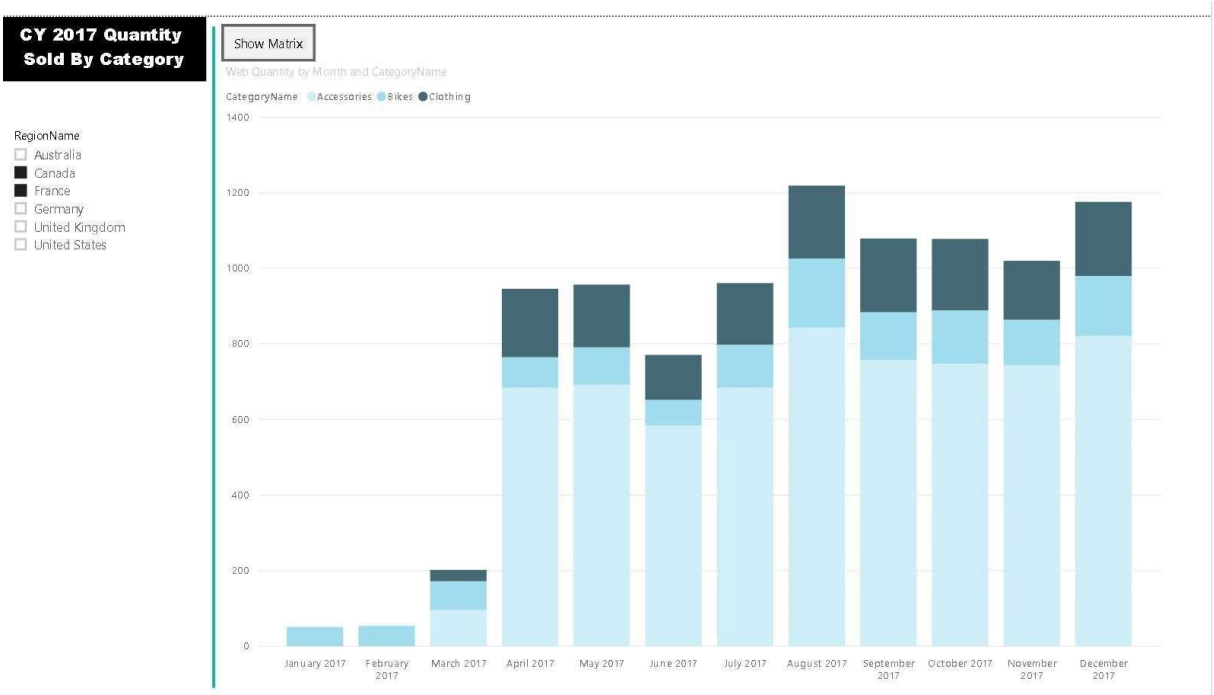
   - **Properties | Position** ○

     Horizontal = 240 ○

     Vertical = 100

Exercise 6 Test the Bookmarks

1. Click the **'X'** to close both the **Selection** and **Bookmarks** panes.
2. Hold down the **CTRL key** and test the Bookmark buttons. They should toggle the relevant chart on and off.
3. **Save** the Power BI Desktop file.
4. Close Power BI Desktop.



Result: You have created some bookmark buttons, that when clicked, will display the relevant bookmark objects.

Lab: 18 Adding Custom Tooltips

In the following lab, you will create a custom style Tooltip that will display additional Subcategory Sales data when you hover over a subcategory column in the Table or one of the Column charts.

Exercise 1 – Add a Custom Tooltip page

1. On the Windows Taskbar, open the Windows **File Explorer**.
2. Navigate to the **C:\Coursefiles\QAPBIDESK\lab18** folder. Double-click the **QAPBID Lab 18 Start.pbix** file to open it.
3. Select **Report** view.
4. **Add** a **new page**. Name the page **Subcategory Tooltip**.
5. On the **Visualizations** pane, click the **Format** icon and set the following:
   - **Page Information** ○ **Allow use as tooltip** = On
   - **Canvas settings** ○ **Type** = Custom

     Height = 450 px

     Width = 300 px
6. At the top of the page, click the **View** menu. On the ribbon, click the **Page view** icon and ensure that **Actual size** is selected. This will display the actual layout size.
7. On the **Visualizations** pane, to the left of the **Format** icon, click the **Build visual** icon (Add data to your visual).
8. From the **Fields** pane, select the **Product** table. Drag the **SubcategoryName** field onto the **Tooltip bucket (Drag tooltip fields here)**, which is on the **Visualizations** pane. Select **Touring Bikes** in the filter list.
9. Drag the **SubcategoryName** field onto the Report canvas.
10. Change the default **Table** visual to a **Card** visual.
11. Click the **Format visual** icon. Set the **Category label** to **Off**.
12. **Resize** the **Card visual** to be the width of the page with only with necessary height to display the 'Touring Bikes' text. Position it at the top then deselect it.
13. **Add** a **Multi-row card** visual to the page. From the **Fields** pane, select the following fields to populate the Multi-row card:
    - **Web Sales ** Web Quantity
    - **Web Sales ** Web Sales Amount
    - **Web Sales ** Web Margin
    - **Web Sales ** Web Margin %

14. **Resize** the **Multi-row card** visual to display the four values without wasting space. Click the **Format visual** icon and under **Callout values**, adjust the font size if necessary. Place the Multi-row card visual just above the mid-point of the page and then deselect it.

15. Add a **Donut Chart** to the bottom section of the page. Drag the following fields into their respective bucket to populate the **Donut Chart:**

    - **Product ** Size to **Legend**
    - **Web Sales** \ Web Quantity to **Values**

16. On the **Page Tab** at the bottom of the report canvas, **right-click** the **Subcategory Tooltip** page tab and click **Hide Page**. This page will be hidden if published to the Power BI Service.

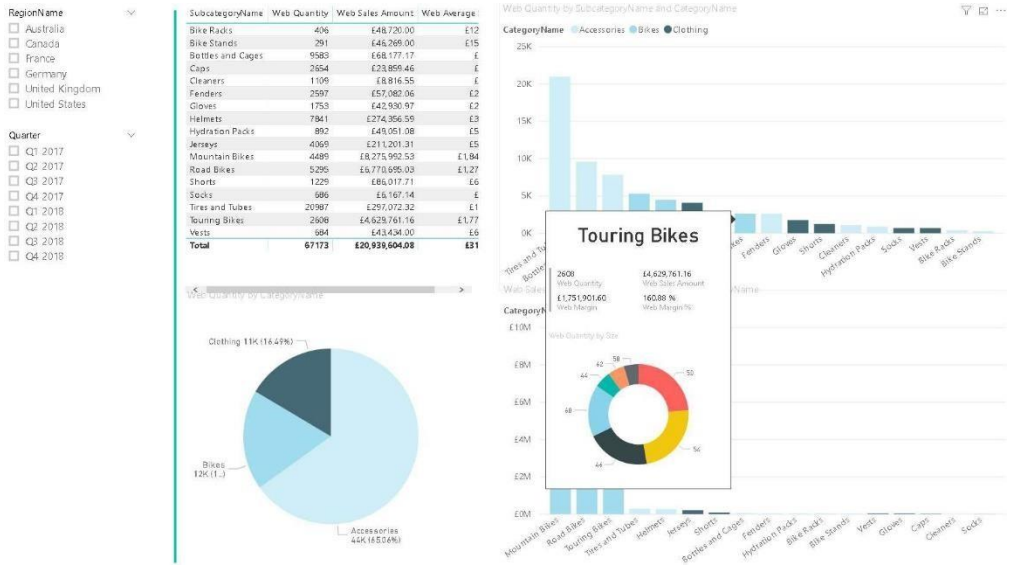
Exercise 2 – Test the Tooltip

1. **Select** the **Regional Report** page.

2. **Hover over** any of the subcategory **columns** in the **Column charts**. The Custom Tooltip will display for the selected subcategory.

3. **Save** the Power BI Desktop file.

4. Close Power BI Desktop.



Result: You have created a Custom Tooltip that displays additional Subcategory Sales data when you hover over a subcategory column in the Table or Column charts.